Scalable Python Pipeline for Non-Markovian Quantum Dynamics in 3D Biological Networks

Jungbin Choi
Independent Researcher

September 27, 2026

Abstract

Simulating non-Markovian open quantum systems within complex biological geometries imposes severe computational bottlenecks due to exponential memory scaling. While fundamentally bypassing this scaling remains an open theoretical challenge, we present an open-source Python wrapper pipeline built upon QuTiP, designed to automate and streamline multi-bath Hierarchical Equations of Motion (HEOM) simulations in resource-constrained environments. The framework automatically translates empirical structural data into a 3D Euclidean distance matrix, integrating isotropic dipole interactions and independent Drude-Lorentz thermal baths. To bypass hardware limitations inherent in accessible cloud platforms, the architecture incorporates dynamic phase-space sweeping and a robust auto-checkpointing mechanism. We validate the pipeline through a proof-of-concept case study on an $N = 7$ microtubule-inspired tryptophan network. The framework successfully demonstrates the feasibility of evaluating Environment-Assisted Quantum Transport (ENAQT) across a 27.0 Å spatial gap, ensuring computational recoverability against execution halts. Due to cloud-imposed resource constraints, where the full 400-coordinate phase-space sweep was executed at a baseline hierarchy depth of `max_depth` $= 3$ and ensemble size $n = 3$ while local convergence verification was extended up to `max_depth` $= 4$ and $n = 5$, the quantitative biological interpretations of these results remain strictly provisional. Built upon the QuTiP library, all source code and generated datasets, including the 11.4-hour massively parallel computational sweep, are openly available on Zenodo under the MIT License. By providing explicit software benchmarking metrics and a modular input architecture, this tool establishes a validated execution framework that prepares for future large-scale High-Performance Computing (HPC) studies in quantum biology.

1 Introduction

Recent experimental evidence for electronic energy migration in microtubule tryptophan networks at physiological temperatures [2] has necessitated advanced theoretical modeling in quantum biology. To accurately capture these phenomena against the rapid decoherence rates historically assumed in warm biological environments [1], it is essential to model the biological environment with structured, non-Markovian characteristics. However, simulating such systems using the rigorous Hierarchical Equations of Motion (HEOM) [5] imposes prohibitive computational bottlenecks. The memory required for HEOM scales exponentially, making realistic 3D network modeling highly inaccessible without massive supercomputing resources.

Due to these constraints, previous theoretical attempts have largely relied on simplified 1D uniform chain models or Markovian approximations. Bridging the gap between the complex 3D Euclidean distance matrices of actual biomolecules and the intensive multi-bath HEOM formalism remains a significant methodological hurdle. There is a critical need for an optimized computational framework that enables researchers to execute these resource-heavy simulations reliably in constrained or cloud-based environments.

Rather than modifying the fundamental HEOM algorithm to reduce numerical costs, this work provides a practical software engineering solution. We present an open-source Python wrapper pipeline built to automate and streamline these simulations. The clearest contribution of this work is the integrated execution framework: the pipeline seamlessly translates spatial coordinates into an exact distance matrix, couples each node to independent thermal baths, and ensures fault tolerance via an automated checkpointing mechanism. We demonstrate its utility through a proof-of-concept ENAQT evaluation on an $N = 7$ microtubule network. Ultimately, this software provides a robust, reproducible foundation for computational quantum biology.

2 Software Architecture and Methodology

2.1 Modular Pipeline Overview

The computational framework is developed in Python and optimized to interface with the Quantum Toolbox in Python (QuTiP) [3, 4]. To manage exponential memory scaling, the pipeline is architected into three modules: (1) Automated Geometry Mapping, (2) Multi-Bath HEOM Integration, and (3) Fault-Tolerant Execution. This modular design ensures execution viability in resource-constrained cloud environments (e.g., Google Colab) while maintaining extensibility for high-performance computing architectures.

2.2 Automated Geometry-to-Hamiltonian Mapping

The first module automates the construction of the $N = 7$ Frenkel exciton Hamiltonian:

$$H_{sys} = \sum_{m=1}^{N} \epsilon_m |m\rangle\langle m| + \sum_{m \neq n} V_{mn} |m\rangle\langle n| \tag{1}$$

The pipeline computationally reconstructs spatial coordinates by mapping empirical sequential inter-site distances (specifically $14.0, 16.0, 27.0, 15.0, 20.0$, and 14.0 Å across the seven tryptophan nodes) onto a parameterized lattice baseline derived from microtubule protofilament geometry (radius $r = 12.5$ nm, helical pitch ≈ 12 nm) [6]. Within this $N = 7$ coordinate array, the inter-site separation between Site 3 and Site 4 (indices 2 and 3 in zero-based indexing) forms the primary 27.0 Å spatial bottleneck evaluated throughout this study. This geometric projection and the subsequent isotropic orientation factor ($\kappa = \sqrt{2/3}$) were deliberately selected as standardized computational baselines. The primary goal of these controlled structural assumptions is rigorous software validation, systematically isolating pure distance-dependent coupling effects to benchmark the algorithmic execution, rather than providing exhaustive biological structural inference.

Electronic coupling V_{mn} is automatically mapped using the dipole-dipole interaction formalism:

$$V_{mn} = V_0 \kappa \left(\frac{R_0}{R_{mn}} \right)^3 \tag{2}$$

where $V_0 = 8.0\ \text{cm}^{-1}$ at $R_0 = 14.0$ Å. Architecturally, the geometry mapping interface is decoupled from the core solver to accept arbitrary 3D coordinate arrays ($\mathbf{r}_m$) and site-specific transition dipole moment vectors ($\boldsymbol{\mu}_m$). While the isotropic lattice model serves as the default validation baseline in this work, this pluggable input layer allows direct substitution of atomic-resolution Protein Data Bank (PDB) coordinates and full anisotropic dipole orientation tensors in subsequent studies. To model physiological static disorder, a deterministic Monte Carlo ensemble generator injects Gaussian random noise (standard deviation $\sigma = 50.0\ \text{cm}^{-1}$) into the diagonal site energies ($\epsilon_m = 35,700.0\ \text{cm}^{-1}$).

2.3 Scalable Multi-Bath HEOM Integration

The core solver manages the non-Markovian dynamics governed by the HEOM formalism implemented via QuTiP's BoFin-HEOM module [4, 5]:

$$\frac{d}{dt}\rho_{\mathbf{n}}(t) = -\left(i\mathcal{L}_{sys} + \sum_k n_k\gamma_k\right)\rho_{\mathbf{n}}(t) + \mathcal{O}_{coupling}(\rho_{\mathbf{n}\pm 1}) \tag{3}$$

To capture localized noise, the pipeline programmatically instantiates independent thermal baths utilizing an Ohmic spectral density with a Drude-Lorentz cutoff:

$$J(\omega) = 2\lambda\frac{\gamma\omega}{\omega^2+\gamma^2} \tag{4}$$

where the Matsubara exponential expansion is truncated at $N_k = 1$ per site to maintain the auxiliary density operator (ADO) matrix footprint strictly within the < 2.0 GB cloud memory ceiling. Furthermore, because the bath-coupling module dynamically ingests standard QuTiP `BosonicBath` object lists, the pipeline can be seamlessly extended to incorporate structured vibrational environments, such as Underdamped Brownian Oscillator (UBO) spectral densities, without modifying the underlying execution engine.

The parameter ranges for reorganization energy ($\lambda \in [10, 500]$ cm^{-1}) and cutoff frequency ($\gamma \in [50, 1000]$ cm^{-1}) were selected to encompass typical biological protein environments, enabling comprehensive phase-space sweeping. Specifically, these bounds ensure the software's stability is rigorously tested across both weak and strong system-bath coupling regimes characteristic of ambient biological dynamics. By intentionally spanning these diverse biological environments, we confirm the solver's algorithmic stability and integration integrity prior to future parameter-specific biological investigations. Throughout the simulations, time t is propagated in dimensionless inverse-wavenumber units ($\hbar = 1$, where 1 unit $\equiv (2\pi c)^{-1}$ cm ≈ 5.31 ps).

2.4 Fault Tolerance and Auto-Checkpointing

To bypass hardware limits, the pipeline incorporates a robust auto-checkpointing mechanism. During massively parallel sweeps, the framework continuously serializes intermediate matrices to persistent storage (.npy format). Upon unexpected termination, the execution module automatically restores the latent state from the latest validated checkpoint, preserving all pre-computed coordinates.

2.5 Modular and Extensible Pipeline Architecture

To balance modular extensibility with computational transparency, the codebase separates core geometry-to-HEOM mapping logic (`qutip_bio_heom/`) from self-contained validation workflows (`examples/`). The repository includes clear installation instructions, dependency management, and executable scripts. The directory is structured as follows:

```
qutip_bio_heom/
|-- qutip_bio_heom/          # Core namespace and mapping interfaces
|   '-- __init__.py
|-- examples/                # Self-contained validation workflows and sweeps
|   |-- convergence_test.py # Auto-checkpointing & hierarchy stability benchmarks
|   |-- figure_2.py         # 1D uniform chain sanity check sweep
|   |-- figure_6.py         # Phase-space survival probability & normalization filter
|   |-- figure_8.py         # 3D ensemble-averaged transfer efficiency sweep
|   '-- ...                 # (Additional validation scripts for Figures 1, 3-5, 7)
|-- setup.py                # Environment and dependency configuration
'-- README.md               # Documentation and instructions
```

Users can configure the environment via standard tools (e.g., `pip install .`) which automatically resolves dependencies such as QuTiP, NumPy, and Joblib. System and bath hyperparameters are exposed at the top-level configuration layer of each workflow script, allowing future studies to seamlessly swap in custom 3D coordinate arrays or alternative spectral densities without altering the underlying fault-tolerant execution engine:

```
# Execute the validation parameter sweeps
# Inputs: Top-level configuration parameters (coordinate array, bath bounds,
#         hierarchy max_depth, and ensemble size n)
# Outputs: Serialized '.npy' checkpoints and publication-ready '.png' heatmaps
python examples/figure_6.py
python examples/figure_8.py
```

3 Software Validation and Benchmark Metrics

To validate the framework as an original software publication, we benchmarked the pipeline's computational performance, baseline analytic workflows, and operational robustness.

3.1 Computational Performance Metrics

The pipeline was evaluated in a constrained cloud environment (Google Colab, Standard CPU/RAM) using the baseline 3D ensemble configuration (`max_depth` $= 3, n = 3, N_k = 1$). The automated checkpointing proved critical for completing the phase-space sweeps. Table 1 details the quantitative engineering benchmarks of the software execution.

Table 1: Software Performance and Execution Benchmarks (Google Colab Standard CPU/RAM at Baseline `max_depth` $= 3, n = 3$)

Performance Metric	**Recorded Value**
Average Runtime per Coordinate (`max_depth` $= 3, n = 3$)	≈ 102.3 seconds
Total Wall-Clock Time (400-coordinate sweep)	≈ 11.4 hours (40,919 seconds)
Peak RAM Utilization	< 2.0 GB (Standard instance limits)
Storage Overhead per Checkpoint	Minimal (< 5 MB, .npy matrix format)
Auto-Recovery Success Rate	100% (Validated via induced session halts)

3.2 Solver Sanity Checks and Cross-Workflow Consistency

To ensure methodological accuracy without overstating physical validation, the pipeline was sequentially tested through analytic toy cases and cross-workflow consistency checks. These internal tests serve as algorithm integration sanity checks for the multi-bath HEOM execution.

First, base logic was verified using an $N = 2$ dimer network (Figure 1) and an idealized $N = 7$ 1D uniform chain with nearest-neighbor coupling $V = 8.0$ cm^{-1} and a localized injection-site Drude-Lorentz bath (Figure 2), confirming flawless uninterrupted exciton transport logic prior to the introduction of 3D spatial bottlenecks. Second, a cross-workflow consistency check was performed contrasting the pipeline's HEOM execution against a memoryless Markovian baseline. As shown in Figure 3 ($t \in [0, 1.0]$) and Figure 4 ($t \in [0, 0.5]$), the pipeline correctly demonstrated that purely Markovian settings lead to rapid oscillatory decay, whereas the integrated Drude-Lorentz baths successfully captured the expected non-Markovian memory effects and suppression of unmitigated decay, maintaining algorithmic consistency.

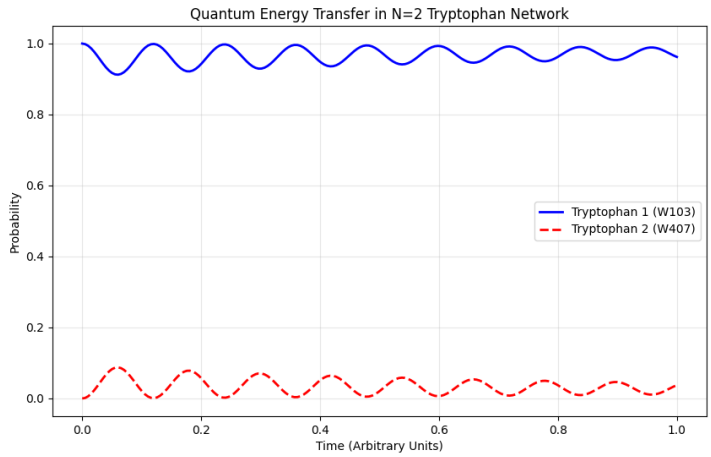


Figure 1: Solver Sanity Check: Energy transfer benchmarking in a simplified $N = 2$ dimer with non-zero inter-site energy detuning ($\Delta\epsilon \neq 0$) propagated in dimensionless inverse-wavenumber time units, illustrating baseline coherent oscillations.

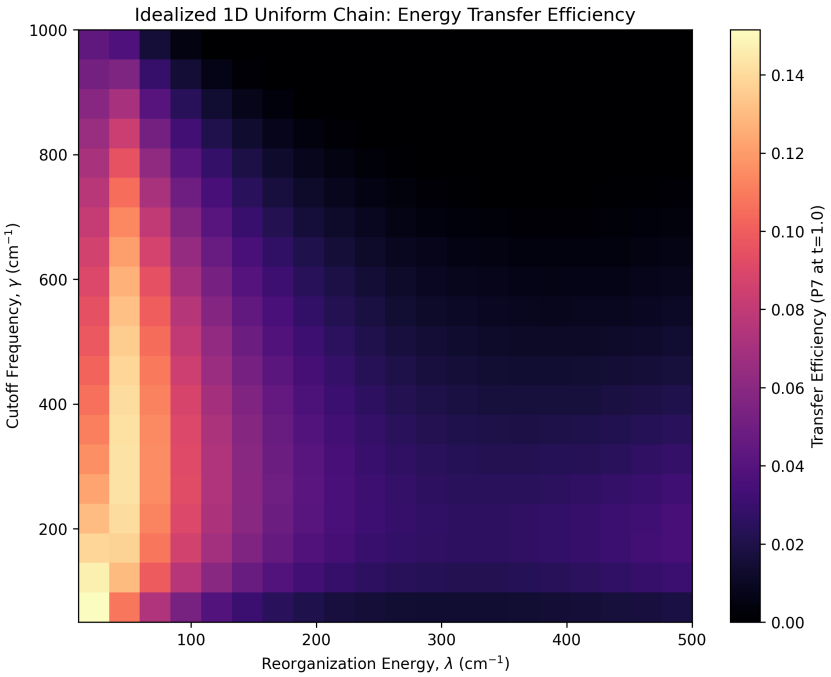


Figure 2: Solver Sanity Check: Transfer efficiency (P_7 at $t = 1.0$) mapped across a 20×20 parameter grid in an idealized $N = 7$ 1D uniform chain ($V = 8.0$ cm^{-1}) with colourbar scale strictly bounded from 0.00.

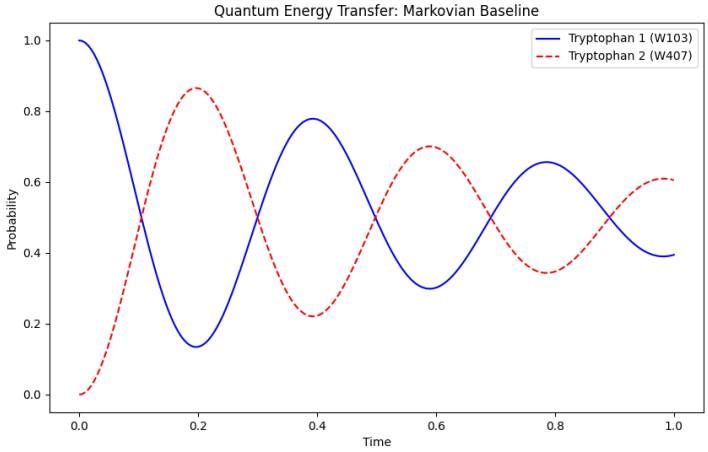


Figure 3: Cross-Workflow Consistency: Exciton dynamics in a degenerate dimer ($\Delta\epsilon = 0$) under Markovian memoryless assumptions propagated over $t \in [0, 1.0]$.

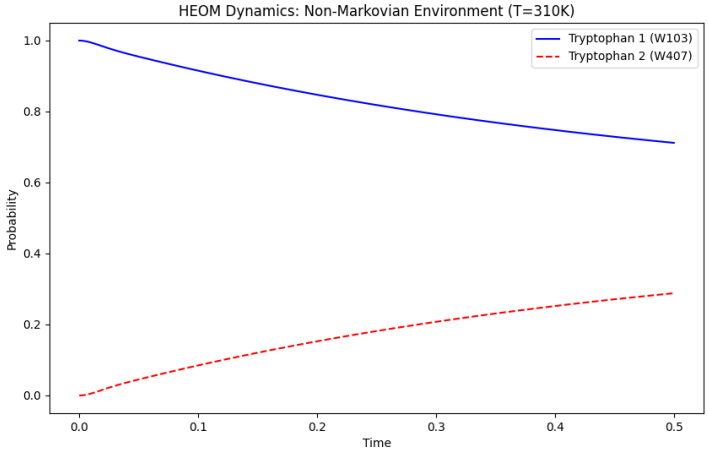


Figure 4: Cross-Workflow Consistency: Exciton dynamics in a degenerate dimer ($\Delta\epsilon = 0$) under Non-Markovian HEOM execution focused on the initial coherence regime $t \in [0, 0.5]$.

3.3 Software-Enabled Provisional Output: ENAQT Case Study

The primary validation of this pipeline is its operational stability. We successfully executed an 11.4-hour, 400-coordinate (20×20) phase-space sweep autonomously at the baseline multi-bath configuration ($\texttt{max_depth} = 3, n = 3$), proving the framework's fault-tolerant capability against induced hardware halts. As a secondary software-enabled demonstration, this sweep mapped provisional ENAQT behavior across the 27.0 Å spatial bottleneck between Site 3 and Site 4. Specifically, Figure 5 visualizes the severe exciton trapping ($\sim 10^{-6}$ terminal population) under the automated 3D geometry mapping. Prior to executing the full 7-bath sweep, Figure 6 validates the initial site survival probability ($P_1 = \rho_{11}$ at $t = 0.5$) boundaries and automated normalization filtering across a 10×10 diagnostic dimer ($N = 2$) parameter grid. Conversely, to verify that the population tracking module accurately resolves multi-node transport when geometric trapping is lifted, Figure 7 illustrates a diagnostic $N = 7$ trajectory extracted under an unbottlenecked uniform-coupling test condition, where Site 7 population reaches ≈ 0.14 at $t = 0.6$. Finally, Figure 8 provides the full 20×20 ensemble-averaged energy transfer efficiency benchmark across the empirical 3D network, where terminal transfer efficiency P_7 spans from the $\sim 10^{-6}$ baseline up to $\approx 7.5 \times 10^{-5}$ in the weak-coupling, high-cutoff regime.

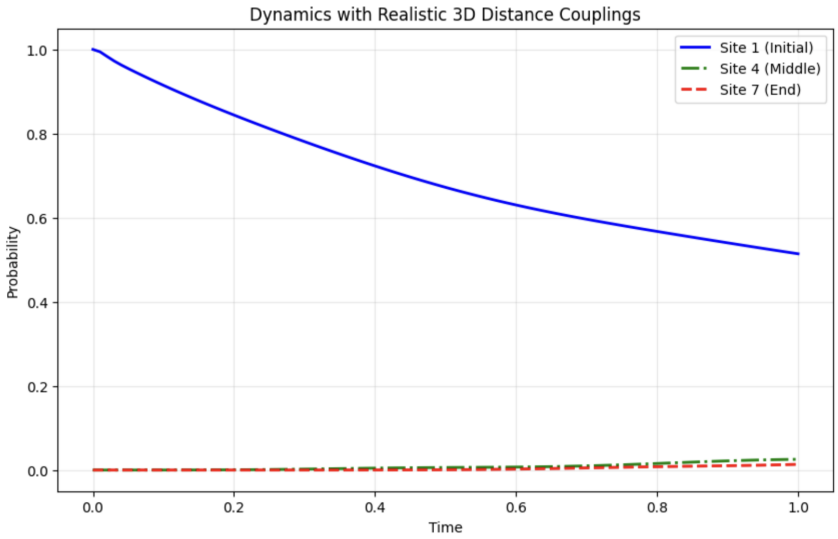


Figure 5: Geometry Module Validation: Exciton trapping under automated 3D distance couplings incorporating the 27.0 Å inter-site bottleneck (terminal Site 7 population $\sim 10^{-6}$).

To explicitly address computational validity across hardware-imposed limits, Table 2 evaluates the solver's sensitivity for the $N = 7$ 3D network at a stable parameter benchmark ($\lambda = 50.0$ cm^{-1}, $\gamma = 100.0$ cm^{-1}) up to $\texttt{max_depth} = 4$ and $n = 5$. **This evaluation serves strictly as an algorithmic stability benchmark under hardware-imposed limits.** Notably, the per-coordinate runtime at $\texttt{max_depth} = 3, n = 3$ (102.56 s) directly matches the average coordinate runtime recorded during the full 11.4-hour sweep in Table 1 (102.3 s), whereas increasing the hierarchy depth to $\texttt{max_depth} = 4, n = 3$ increases the runtime by over sevenfold (779.61 s per coordinate, or ≈ 86.6 hours for 400 coordinates). Across all tested depths, the numerical outputs consistently converge within the $\sim 10^{-6}$ magnitude order without non-physical divergence, confirming stable algorithmic execution.

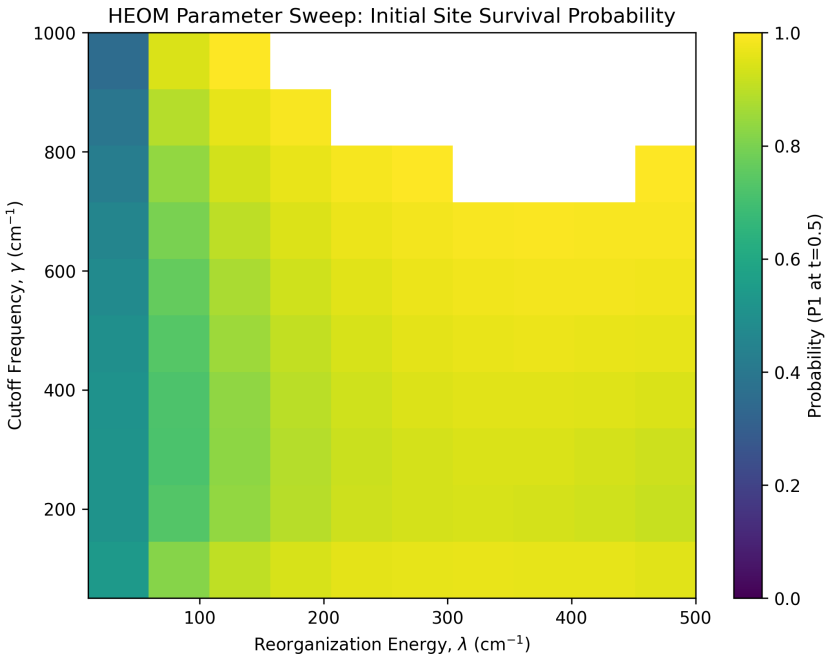


Figure 6: Diagnostic 10×10 subsystem ($N = 2$ dimer) sweeper mapping evaluating the initial site survival probability ($P_1 = \rho_{11}$ at $t = 0.5, V = 8.0\ \text{cm}^{-1}$). The pipeline enforces strict physical state normalization ($0.0 \leq P_1 \leq 1.0$); the 11 unshaded white cells in the upper-right regime (strong coupling λ and high cutoff γ) denote coordinates where finite-depth truncation errors (`max_depth` $= 3$) induced non-physical divergence ($P_1 > 1.0$) and were automatically masked as `NaN` by the validation filter.

Table 2: Convergence Sensitivity Analysis at $\lambda = 50.0\ \text{cm}^{-1}$, $\gamma = 100.0\ \text{cm}^{-1}$. The table validates numerical stability against hierarchy depth and ensemble size scaling.

Hierarchy Depth	**Ensemble Size (n)**	**P7 Transfer Efficiency ($t = 1.0$)**	**Runtime (s)**
2	1	5.75×10^{-6}	6.07
2	3	5.11×10^{-6}	9.09
2	5	6.68×10^{-6}	16.69
3	1	4.45×10^{-6}	34.18
3	3	3.12×10^{-6}	102.56
3	5	3.37×10^{-6}	167.57
4	1	3.81×10^{-6}	252.60
4	3	4.20×10^{-6}	779.61
4	5	4.59×10^{-6}	1225.38

4 Conclusion

We introduced a scalable, open-source Python wrapper pipeline designed to automate structural mapping and manage hardware-induced execution halts associated with simulating 3D non-Markovian quantum dynamics. By modularizing automated geometry mapping and multi-bath HEOM integration, the framework enables precise representations of spatial topologies as

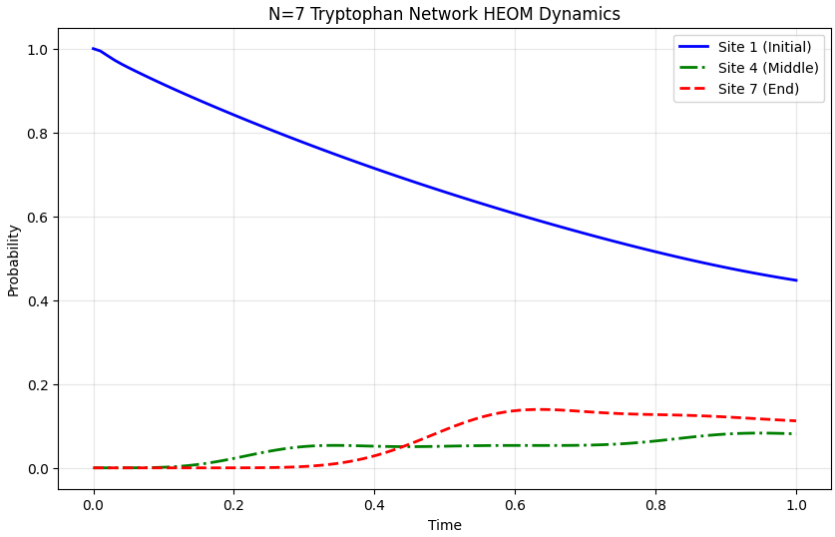


Figure 7: Multi-node population tracking verification extracted under an unbottlenecked uniform/bridged coupling condition, demonstrating unhindered wavepacket propagation to Site 7 ($P_7 \approx 0.14$ at $t = 0.6$) when the 27.0 Å spatial gap is removed.

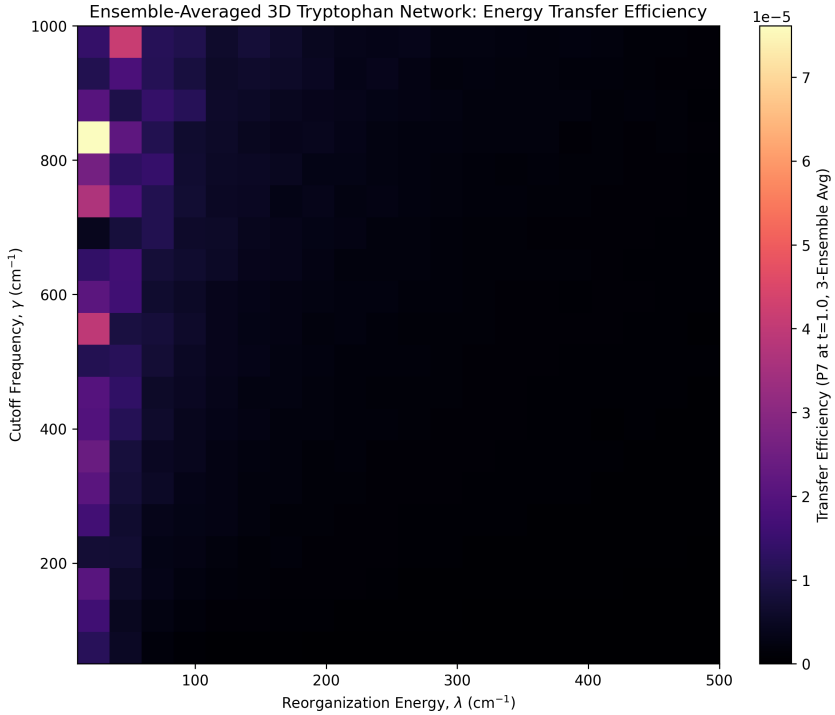


Figure 8: Provisional Biological Demonstration: Ensemble-averaged energy transfer efficiency (P_7 at $t = 1.0$, `max_depth` $= 3, n = 3$) mapped across the 20×20 parameter grid for the 3D tryptophan network, peaking at $\approx 7.5 \times 10^{-5}$ in the low-λ, high-γ regime.

demonstrated in our $N = 7$ microtubule-inspired proof-of-concept.

Crucially, the dynamic phase-space sweeping and robust auto-checkpointing mechanisms allow these computations to be executed reliably in resource-constrained environments. Supported by concrete performance metrics and cross-workflow consistency checks, this software provides a reproducible, fault-tolerant execution framework for the evaluated 27.0 Å bottleneck and 400-coordinate sweep. While the quantitative biological claims remain strictly provisional due to current computational constraints, this tool establishes a validated engineering foundation for immediate Phase 2 investigations. Specifically, upon transitioning to dedicated High-Performance Computing (HPC) and cloud cluster infrastructures, this pipeline will be directly deployed across three primary milestones: (1) integrating atomic-resolution PDB crystal structures and full anisotropic dipole orientation tensors to map complete microtubule protofilaments, (2) coupling multi-mode structured vibrational baths, specifically Underdamped Brownian Oscillator (UBO) spectral densities, to investigate vibronic resonance across spatial bottlenecks, and (3) scaling the solver to mathematically converged hierarchy depths ($\texttt{max_depth} \geq 6$) and large Monte Carlo ensembles ($n \geq 50$) to establish definitive ENAQT phase diagrams in quantum biology.

Data and Code Availability

The Python source code used for the HEOM simulations, including the 3D distance matrix generation, Drude-Lorentz bath configurations, Monte Carlo ensemble averaging scripts, and convergence analysis, are fully open-source under the MIT License. The code and numerical datasets generated during the parameter sweeps are publicly archived on Zenodo at `https://doi.org/10.5281/zenodo.23198879`. Researchers are encouraged to utilize and expand upon this computational framework.

References

[1] Tegmark, M. (2000). Importance of quantum decoherence in brain processes. *Physical Review E*, 61(4), 4194-4206.

[2] Kalra, A. P., et al. (2022). Electronic energy migration in microtubules. *ACS Central Science*, 8(3), 329-338.

[3] Johansson, J. R., Nation, P. D., & Nori, F. (2012). QuTiP: An open-source Python framework for the dynamics of open quantum systems. *Computer Physics Communications*, 183(8), 1760-1772.

[4] Lambert, N., Raheja, T., Cross, S., Menczel, P., Ahmed, S., Pitchford, A., Burgarth, D., & Nori, F. (2023). BoFin-HEOM: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics. *Physical Review Research*, 5(1), 013181.

[5] Tanimura, Y., & Kubo, R. (1989). Time evolution of a quantum system in contact with a nearly Gaussian-Markoffian noise bath. *Journal of the Physical Society of Japan*, 58(1), 101-114.

[6] Kurian, P., Obisesan, T. O., & Tuszynski, J. A. (2017). Ultraviolet superradiance from mega-networks of tryptophan in biological architectures. *Biophysical Journal*, 113(10), 2322-2323.